FRAUD RISK INTELLIGENCE

Model Card

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1. MODEL OVERVIEW

1.1 Model Description

The Fraud Risk Intelligence Model is a machine learning-based classification system developed to estimate the probability that a financial transaction is fraudulent.

The model analyzes transaction characteristics, customer-related information, behavioral indicators, authentication details, and risk scores to identify potentially suspicious financial activities.

The framework combines predictive modeling with probability calibration, cost-sensitive threshold optimization, explainable artificial intelligence, and operational decision simulation.

Unlike a conventional binary fraud classifier, the proposed framework supports three transaction decisions:

- **APPROVE:** Transactions with relatively low estimated fraud risk.
- **MANUAL REVIEW:** Transactions requiring additional investigation.
- **PROPOSE BLOCK:** Transactions identified as sufficiently high-risk to warrant consideration for blocking, subject to appropriate human oversight.

The objective is to support fraud investigation and risk-based decision-making while considering financial losses, false positives, operational workload, and customer impact.

The system is designed as a research prototype and has not been approved for deployment in a live financial environment.

1.2 Model Identification

Property	**Description**
Model Name	Fraud Risk Intelligence
Model Category	Supervised Binary Classification
Base Algorithm	Logistic Regression
Class Imbalance Strategy	Unweighted
Probability Calibration	Sigmoid Calibration
Input Features	23 Original Predictive Features

Target Variable	fraud
Prediction Output	Estimated Fraud Probability
Decision Output	Approve / Manual Review / Proposed Block
Explainability Method	SHAP
Implementation Language	Python
Deployment Status	Research Only

1.3 Model Version Identification

The selected model was frozen before final testing.

A SHA-256 checksum was generated to identify the model artifact and support reproducibility.

Model SHA-256:

87217ee986fbed6763df7111294c051c94fb7934ea9ce3e185064c6841bce517

The checksum can be used to verify that the model artifact has not been unintentionally modified after evaluation.

2. INTENDED USE AND RESTRICTIONS

2.1 Intended Use

The model is intended for research, academic evaluation, and controlled fraud risk analysis.

Potential applications include:

1. Estimating transaction-level fraud probabilities.
2. Supporting fraud investigation prioritization.
3. Comparing alternative transaction decision policies.
4. Simulating manual review workloads.
5. Evaluating hypothetical financial consequences of fraud decisions.
6. Investigating the effects of model calibration and class imbalance.
7. Supporting model explainability and governance research.

The model may also be used to demonstrate how machine learning outputs can be integrated into an operational decision-support framework.

2.2 Out-of-Scope Uses

The model is not intended for:

- Fully automated blocking of live financial transactions.
- Unsupervised customer account suspension.
- Making irreversible financial decisions without human oversight.
- Production deployment without independent validation.
- Estimating verified financial savings from hypothetical costs.
- Establishing fraudulent intent solely from model predictions.

The model has not undergone sufficient independent validation, fairness assessment, or operational approval for these purposes.

2.3 Human Oversight

The framework requires human oversight for customer-impacting decisions.

High-risk classifications should be interpreted as indicators requiring additional investigation rather than conclusive evidence of fraud.

Any future operational implementation should include human approval procedures, decision logging, customer appeal mechanisms, and appropriate override controls.

3. DATASET INFORMATION

3.1 Dataset Description

The model was developed using a financial transaction dataset containing 3,000 observations and 25 original columns.

Each observation represented an individual transaction.

The dataset included information relating to transaction amounts, transaction frequency, customer account characteristics, device behavior, authentication, and risk-related indicators.

The target variable was fraud, where:

0 = Legitimate Transaction

1 = Fraudulent Transaction

Table 3.1 Dataset Overview

Attribute	Value
Total Transactions	3,000
Original Columns	25
Legitimate Transactions	2,636
Fraudulent Transactions	364
Fraud Prevalence	12.13%
Total Missing Values	582
Transactions with Missing Values	542
Duplicate Records	0
Duplicate Transaction IDs	0

The class distribution indicated that legitimate transactions were considerably more common than fraudulent transactions.

Therefore, specialized evaluation metrics were required to assess fraud detection performance.

3.2 Dataset Partitioning

The dataset was divided using stratified random sampling.

Table 3.2 Training, Validation and Testing Datasets

Dataset	Transactions	Fraud Cases	Fraud Rate
Training	1,800	218	12.11%
Validation	600	73	12.17%
Final Test	600	73	12.17%

The training dataset was used for model development and cross-validation.

The validation dataset supported model comparison, calibration assessment, and decision policy selection.

The final testing dataset was reserved for evaluating the frozen model and decision thresholds.

The final holdout evaluation was performed without retraining or modifying the selected decision policy.

3.3 Data Quality Considerations

A data quality investigation identified missing values, outliers, and potential prediction-time data availability risks.

The authentication_level feature contained 289 missing observations.

An IQR-based investigation identified 565 transactions containing at least one unusual numerical observation.

Outliers were retained because unusual transaction patterns may contain valuable fraud detection information.

Missing values were handled through model preprocessing, while missingness indicators were considered during feature engineering experiments.

Historical and risk-related predictors were also examined for potential data leakage.

Several features still require verification that they would be available before transaction authorization in a real operational environment.

3.4 Data Limitations

The dataset did not provide sufficient information for strict chronological validation or customer-level separation.

Complete transaction timestamps and customer identifiers were unavailable.

Consequently, the evaluation could not establish performance on genuinely future transactions or entirely unseen customer groups.

The original dataset's institutional provenance and representativeness have not been independently established within this project.

These limitations restrict claims about real-world generalization.

4. INPUT FEATURES AND PREPROCESSING

4.1 Original Input Features

The original dataset contained 25 columns.

The transaction identifier and fraud target were excluded from predictive inputs.

Consequently, 23 original features were available for model development.

The features represented several categories.

Transaction Characteristics

- Transaction amount
- Transaction hour
- Day of week
- Transaction channel
- Card presence
- Merchant category

Behavioral Characteristics

- Transactions during the previous hour
- Transactions during the previous 24 hours
- Failed login attempts
- Distance from usual transaction location

Customer and Device Characteristics

- Customer account age
- Device age
- Email age
- Customer tier
- Device type
- New-device indicator

Risk and Authentication Characteristics

- Merchant risk score
- IP risk score
- Authentication level
- Prior chargebacks
- Billing address mismatch

Additional transaction-context features were also retained according to the validated input schema.

4.2 Data Preprocessing

The modeling pipeline incorporated preprocessing procedures appropriate for numerical and categorical variables.

The preprocessing workflow included:

1. Numerical missing-value treatment.
2. Categorical missing-value handling.

3. Categorical encoding.
4. Numerical feature scaling where required.
5. Model fitting and prediction.

Preprocessing was incorporated into the machine learning pipeline to reduce the risk of information leakage during cross-validation.

The pipeline was designed to handle previously unseen categorical values.

4.3 Feature Engineering Experiments

Sixteen additional features were developed during feature engineering experiments.

These included velocity ratios, logarithmic transformations, risk score combinations, missingness indicators, temporal indicators, and interaction terms.

The engineered feature set contained 39 predictors.

However, the engineered Logistic Regression model did not consistently outperform the original-feature model during validation.

Therefore, the final selected model used the original 23-feature configuration.

This decision was based on measured validation performance rather than model complexity.

1. 05. MODEL DEVELOPMENT METHODOLOGY

5.1 Candidate Models

Three machine learning algorithms were evaluated:

1. Logistic Regression
2. Random Forest
3. XGBoost

Logistic Regression provided a transparent baseline.

Random Forest and XGBoost were investigated as nonlinear alternatives capable of learning more complex relationships.

Model training used stratified cross-validation.

Random Forest and XGBoost underwent limited hyperparameter optimization.

5.2 Model Comparison

Table 1.1 Initial Validation Model Performance

Model	PR-AUC	ROC-AUC

Logistic Regression — Original	0.2424	0.6895
Logistic Regression — Engineered	0.2203	0.6821
Random Forest — Engineered	0.2249	0.6474
XGBoost — Engineered	0.2130	0.6837

The original-feature Logistic Regression model achieved the highest validation PR-AUC and ROC-AUC among the evaluated initial model configurations.

This demonstrated that the simpler classification model performed competitively against the investigated nonlinear alternatives.

However, the nonlinear model hyperparameter searches were limited in size.

Therefore, these experiments did not establish that Logistic Regression would outperform all possible advanced model configurations.

5.3 Class Imbalance Handling

The project compared three imbalance-handling approaches:

- Unweighted classification.
- Class-weighted classification.
- Random oversampling.

Class weighting and random oversampling increased recall at the default 0.50 decision threshold.

However, these strategies also produced substantially higher numbers of false-positive classifications.

The experiments demonstrated that class imbalance handling influenced the trade-off between fraud detection and legitimate transaction disruption.

The final model used the unweighted classification strategy in combination with probability calibration and cost-sensitive decision thresholds.

6. PROBABILITY CALIBRATION

6.1 Calibration Methodology

Probability calibration was investigated to improve the reliability of fraud probability estimates.

The project compared:

- Uncalibrated probabilities.

- Sigmoid calibration.
- Isotonic regression.

Calibration performance was evaluated using the Brier Score, Log Loss, Expected Calibration Error, and ranking metrics.

6.2 Selected Calibration Configuration

Table 6.1 Selected Validation Calibration Results

Metric	Value
Calibration Method	Sigmoid
Average Precision	0.2408
ROC-AUC	0.6928
Brier Score	0.1016
Expected Calibration Error	0.0155
Mean Predicted Fraud Probability	12.51%
Observed Fraud Rate	12.17%

The unweighted Logistic Regression model with Sigmoid Calibration achieved the lowest validation Brier Score among the investigated configurations.

This configuration was selected as the provisional probability scoring model.

Calibration improved the operational interpretability of estimated fraud probabilities.

However, calibration results from a single validation dataset do not guarantee reliability across future transaction periods or customer populations.

7. FINAL MODEL PERFORMANCE

7.1 Final Evaluation

The frozen model was evaluated using the reserved 600-transaction testing dataset.

No post-test model fitting, calibration adjustment, or threshold optimization was performed.

Table 7.1 Final Holdout Performance

Metric	Final Test Result

Test Transactions	600
Fraud Cases	73
Fraud Prevalence	12.17%
Average Precision	0.1932
ROC-AUC	0.6640
Brier Score	0.1039
Log Loss	0.3550
Expected Calibration Error	0.0215
Mean Predicted Fraud Probability	11.87%
Observed Fraud Rate	12.17%

The final Average Precision was lower than the validation Average Precision.

The final ROC-AUC also decreased relative to validation performance.

These findings indicated reduced generalization performance on the held-out evaluation data.

Although the model retained some ability to distinguish fraudulent from legitimate transactions, its predictive performance remained limited.

7.2 Model Calibration Interpretation

The final model produced a mean predicted fraud probability of approximately 11.87%.

The observed fraud prevalence was approximately 12.17%.

The difference between these averages was relatively small.

However, agreement between average predicted and observed fraud rates does not guarantee accurate calibration across individual probability ranges.

The final Brier Score and Expected Calibration Error provide additional evidence regarding probability reliability, but further validation is required.

8. COST-SENSITIVE DECISION POLICY

8.1 Business Cost Assumptions

The project developed a hypothetical financial cost framework to evaluate alternative fraud decision policies.

Table 8.1 Illustrative Business Costs

Cost Component	Assumed Value
Fraud Transaction Approved	LKR 20,000
Legitimate Transaction Blocked	LKR 1,500
Manual Review Cost	LKR 150
Assumed Review Fraud Detection Rate	90%
Assumed Review False Rejection Rate	2%
Maximum Review Rate	20%
Maximum Proposed Block Rate	5%

These values represent research assumptions.

They have not been verified against actual financial institution records.

8.2 Frozen Decision Thresholds

The final model used the following decision thresholds.

Table 8.2 Fraud Decision Thresholds

Estimated Fraud Probability	Proposed Action
Below 0.152046	APPROVE
0.152046 to below 0.262003	MANUAL REVIEW
0.262003 and above	PROPOSE BLOCK

The thresholds were selected during the validation stage.

They were frozen before final testing.

The final test results were not used to modify these values.

8.3 Final Operational Results

Table 8.3 Final Test Transaction Routing

Decision	Transactions	Percentage
Approve	474	79%
Manual Review	114	19%
Proposed Block	12	2%

Total	600	100%

The final testing batch remained within the assumed review and proposed block-rate limits.

However, the outcome analysis identified substantial false-negative and false-blocking risks.

Of the 73 fraudulent transactions, 46 were assigned to the approval category.

Eight of the twelve proposed blocks involved legitimate transactions.

These findings represent important limitations for customer-impacting operational decisions.

8.4 Illustrative Financial Performance

Table 8.4 Final Test Cost Simulation

Cost Measure	**Amount**
Approve-All Scenario Cost	LKR 1,460,000
Proposed Policy Scenario Cost	LKR 997,830
Illustrative Difference	LKR 462,170
Illustrative Reduction	31.66%

The results demonstrated potential financial benefits under the selected hypothetical assumptions.

However, these values do not represent verified realized savings.

The analysis depends on assumed fraud losses, investigation costs, blocking costs, and reviewer effectiveness.

9. MODEL EXPLAINABILITY

9.1 Explainability Framework

SHAP was used to investigate the contribution of input features to model predictions.

The explainability analysis included global feature importance and transaction-level explanations.

For the selected calibrated Logistic Regression configuration, SHAP contributions were calculated for the underlying model's log-odds.

These contributions should not be interpreted as exact additive changes to the final calibrated fraud probability.

9.2 Global Feature Importance

Table 9.1 Leading Global SHAP Features

Rank	Feature	Mean Absolute SHAP
1	Merchant Category	0.31824
2	Transactions in Last Hour	0.26547
3	New Device	0.25423
4	Transaction Channel	0.20099
5	Merchant Risk Score	0.17599

Merchant category, transaction velocity, and new-device activity were among the most influential features.

These findings were broadly consistent with exploratory data analysis.

However, feature importance does not establish that a feature causes fraudulent activity.

9.3 Local Explainability

A local explanation was generated for transaction TXN5000873.

The transaction received a calibrated fraud probability of approximately 44.17%.

Under the frozen decision policy, the transaction was categorized as a proposed block.

Transaction amount, new-device status, and merchant risk score contributed positively to the underlying model score.

Other features contributed negatively.

The explanation demonstrated how transaction-level reason codes could support fraud investigation.

However, explanations should not be used as definitive proof of fraudulent intent.

9.4 Explainability Limitations

SHAP explanations depend on the fitted model, preprocessing transformations, and selected reference distribution.

Feature contributions may change when data distributions or model versions change.

Furthermore, correlated features can complicate interpretation.

Therefore, explanations should be reviewed for stability and counterintuitive behavior before operational use.

10. ROBUSTNESS AND STRESS TESTING

10.1 Stress Testing Methodology

The frozen model and decision policy were evaluated under simulated input perturbations.

The investigated scenarios included:

- Increased numerical missingness.
- Missing authentication information.
- Unseen transaction categories.
- Increased transaction amounts.
- Increased transaction distances.
- Increased transaction velocity.
- Noisy risk score inputs.

The original decision thresholds were retained during testing.

10.2 Important Results

Table 10.1 Selected Robustness Results

Scenario	Review Rate	Block Rate	Action Change
Baseline	20.00%	4.00%	0.00%
Authentication Missing 20%	24.50%	4.83%	6.17%
Unseen Channel 10%	20.50%	3.83%	0.83%
Amount Increase 25%	20.50%	4.17%	0.83%
Distance Increase 50%	20.50%	4.17%	0.83%
Velocity Increase +2	33.83%	7.33%	20.50%

The largest operational impact occurred when transaction velocity increased.

The review rate increased to 33.83%, exceeding the assumed 20% operational capacity.

The proposed block rate increased to 7.33%, exceeding the assumed 5% limit.

These findings indicated that the frozen decision policy could become operationally infeasible under changing transaction conditions.

Therefore, capacity-aware controls and human-approved fallback procedures are required before operational deployment.

11. MODEL LIMITATIONS AND KNOWN RISKS

The following limitations were identified during model development and evaluation.

11.1 Predictive Performance Limitations

The final Average Precision was 0.1932.

Although this exceeded the fraud prevalence reference value, the ranking performance remained limited.

The final policy approved 46 of the 73 fraudulent transactions.

Consequently, the framework cannot be relied upon to identify all suspicious activity.

11.2 False-Positive and Customer Impact Risks

The final policy proposed blocking twelve transactions.

Eight of these transactions were legitimate according to the test labels.

Unsupervised blocking could therefore inconvenience legitimate customers and undermine trust.

Appropriate human review and appeal mechanisms are necessary.

11.3 Data Leakage Risks

Several historical and risk-based features require verification that they would be available at the time of prediction.

Using information generated after transaction authorization could result in misleading model performance.

A point-in-time feature availability contract should be established before deployment.

11.4 Generalization Risks

The absence of complete timestamps and customer identifiers prevented chronological and customer-level validation.

The dataset was also relatively small.

The final testing subset contained only 73 fraudulent transactions.

These factors limit confidence in performance stability across future periods and unseen customers.

11.5 Operational Risks

Stress testing demonstrated that transaction behavior changes could exceed manual review and proposed block capacity limits.

The operational policy therefore requires capacity controls and escalation procedures.

11.6 Business Cost Risks

The financial optimization depended on hypothetical costs and assumed reviewer effectiveness.

Actual business outcomes may differ substantially.

Operational decisions should not be based on the simulated financial savings without verifying the underlying assumptions.

11.7 Fairness and Privacy Risks

A comprehensive protected-group fairness assessment was not performed.

Customer impact and appeal procedures were not validated in an operational environment.

Any future implementation should incorporate privacy safeguards, appropriate data minimization, fairness assessments, and customer redress procedures.

12. MODEL MONITORING FRAMEWORK

A proposed monitoring framework was developed to identify model performance degradation, data quality issues, and operational capacity violations.

Table 12.1 Proposed Monitoring Requirements

Monitoring Metric	**Frequency**	**Warning Trigger**
Manual Review Rate	Per Batch	≥ 18%
Proposed Block Rate	Per Batch	≥ 4.5%
Authentication Missingness	Daily	Baseline + 5 Percentage Points
Unseen Category Rate	Daily	> 1%
Numerical Feature PSI	Weekly	≥ 0.10
Brier Score	After Mature Labels	10% Deterioration
Average Precision	After Mature Labels	10% Deterioration

Manual Review Backlog	Per Batch	Any Pending Cases

Critical operational triggers include review rates exceeding 20% and proposed block rates exceeding 5%.

The monitoring thresholds are proposed governance controls and have not been validated in a live operational environment.

12.1 Model Review Triggers

The model should undergo formal review when:

1. Prediction performance deteriorates beyond approved monitoring limits.
2. Data distributions change significantly.
3. Missing-value rates increase unexpectedly.
4. Previously unseen categories become common.
5. Operational review capacity is exceeded.
6. Customer complaints or false-blocking incidents increase.
7. New fraud patterns are identified.
8. Feature availability or data processing contracts change.

12.2 Retraining Requirements

Retraining should not occur automatically without documented evaluation.

Any updated model should undergo training-only validation, calibration assessment, business cost analysis, robustness testing, explainability review, and governance approval.

A new independent evaluation dataset should be used to assess material model or policy changes.

12.3 Model Suspension Conditions

Any future operational implementation should define conditions under which automated decisions are suspended.

Examples include severe data quality failures, material model performance deterioration, operational capacity breaches, and unacceptable customer-impact incidents.

Such events should trigger escalation to responsible human decision-makers.

13. MODEL GOVERNANCE

13.1 Governance Assessment

The project evaluated governance readiness using ten control gates.

At the initial governance checkpoint, two gates passed.

The baseline transaction review and proposed block rates were within the assumed operational limits.

Other controls remained unresolved, including stress capacity stability, prediction-time data verification, chronological validation, real-world business cost verification, fairness assessment, and operational monitoring.

The reserved final holdout evaluation was completed after the governance checkpoint.

However, completion of that evaluation did not resolve the remaining deployment risks.

13.2 Governance Decision

The final model governance disposition is:

RESEARCH ONLY — NO-GO FOR LIVE AUTOMATED BLOCKING

This decision reflects the following concerns:

- Substantial fraudulent transaction approval rate.
- False blocking recommendations affecting legitimate transactions.
- Operational instability under stress.
- Unverified prediction-time feature availability.
- Limited generalization evidence.
- Hypothetical financial cost assumptions.
- Incomplete fairness and customer appeal assessment.
- Absence of validated live monitoring and human authorization.

The model should remain a controlled research prototype.

14. RECOMMENDED IMPROVEMENTS

Before considering an operational pilot, the following improvements are recommended.

Improved Data Collection: Obtain larger datasets containing complete timestamps, customer identifiers, and verified historical behavioral information.

Independent Validation: Conduct chronological, customer-level, and external evaluations.

Predictive Performance: Investigate additional models and behavioral features using appropriately separated development data.

Business Cost Validation: Replace hypothetical costs with verified financial losses and investigation expenses.

Human Oversight: Establish documented review, override, and appeal procedures.

Capacity Management: Introduce operational controls to manage investigation queues and transaction blocking proposals.

Monitoring: Implement data quality, performance, calibration, drift, and operational workload monitoring.

Fairness Assessment: Investigate potentially disproportionate customer impacts using appropriate and lawfully available evaluation information.

All material model or decision policy changes should undergo independent review before deployment consideration.

15. CONCLUSION

The Fraud Risk Intelligence Model demonstrates how machine learning can be integrated with probability calibration, cost-sensitive threshold optimization, explainability, operational simulation, and model governance.

The final model achieved an Average Precision of 0.1932 and ROC-AUC of 0.6640 on the reserved testing dataset.

The proposed transaction decision policy demonstrated potential financial benefits under hypothetical business cost assumptions.

However, the final evaluation identified substantial limitations involving missed fraudulent transactions, false blocking recommendations, and operational instability.

These limitations prevent the current model from being considered sufficiently reliable for live automated financial decisions.

The framework is therefore recommended for continued research, controlled evaluation, and development of decision-support capabilities.

Final Deployment Recommendation: NO-GO — Research Prototype Only.

APPENDIX A — REPRODUCIBILITY INFORMATION

Component	Details
Development Language	Python

Project Repository	fraud-risk-intelligence
Main Notebook	01_fraud_risk_analysis.ipynb
Model Development	Logistic Regression
Calibration	Sigmoid
Explainability	SHAP
Training Set	1,800 Transactions
Validation Set	600 Transactions
Final Test Set	600 Transactions
Review Threshold	0.1520464035
Block Threshold	0.2620026706
Model Status	Frozen Research Candidate

The project includes reusable Python modules, automated unit tests, technical reports, model evaluation outputs, and a model governance manifest.

The final test evaluation was performed using the frozen model and policy.

The final testing results were not used for model retraining, calibration fitting, or threshold optimization.

APPENDIX B — MODEL CARD DOCUMENT CONTROL

Document Name: Fraud Risk Intelligence Model Card

Required Submission File: 06_model_card.pdf

Associated Project: Fraud Risk Intelligence

Model Status: Research Only

Production Authorization: Not Granted

Final Holdout Evaluation: Completed

Business Cost Validation: Hypothetical Assumptions Only

Monitoring Status: Proposed, Not Operational

Human Approval for Live Blocking: Not Granted